

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Truth table

Row $\leftrightarrow$
Column $\uparrow$

		K-map			
		AB			
C	AB	00	01	11	10
	0	0	1	1	1
1	0	0	0	0	1

$$\overline{B}\overline{C} + A\overline{B} = F$$

$$\overline{0} = 1$$

$$\overline{1} = 0$$

K-map

1, 2, 4, 8, 16, ...

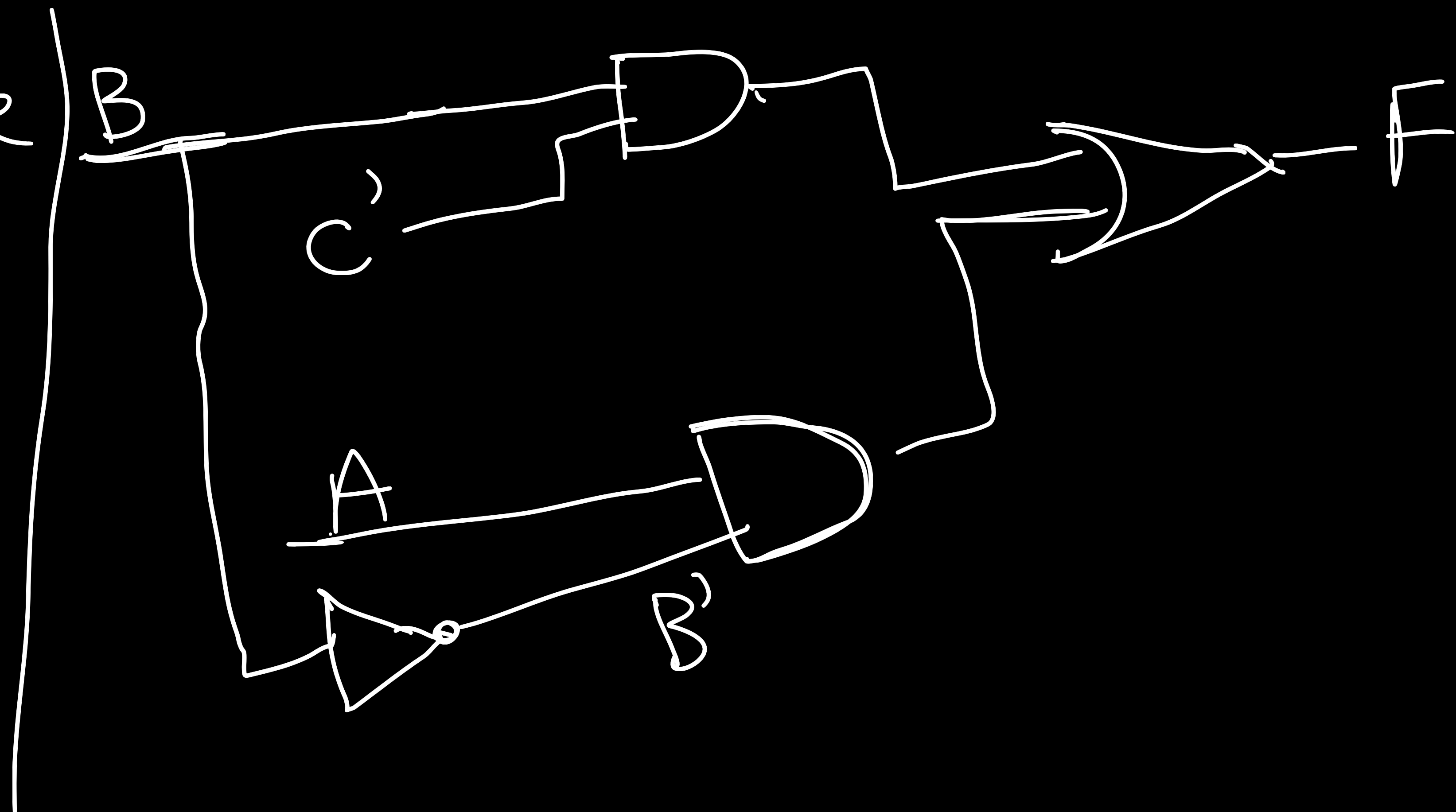
2^n

$n \in [0, \infty)$

Function to circuit

$$F = B\bar{C} + A\bar{B} = BC' + AB'$$

2 AND Gate
1 OR Gate



A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

K-Map

AB

C \ AB	00	01	11	10
0	0	0	1	1
1	1	1	0	1

$$F = \overline{A}C + AC + A\overline{B}C$$

K-Map

C \ AB	00	01	11	10
0	0	1	1	1
1	0	1	0	1

AB

11

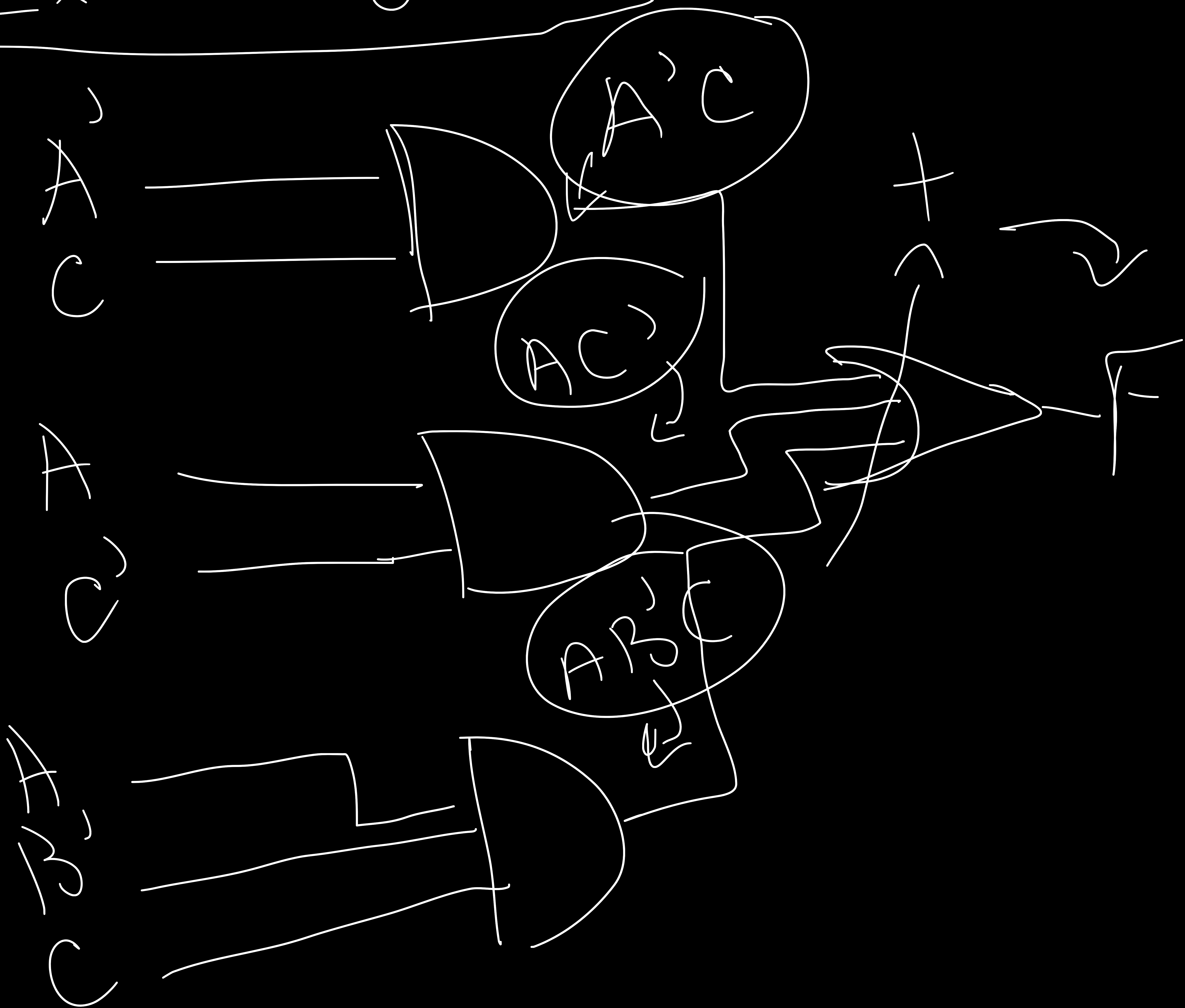
C \ AB	00	01	11	10
0	0	0	1	1
1	1	1	0	1

$$F = \overline{A}C + A\overline{C} + A\overline{B}C$$

$$= A'C + AC' + AB'C$$

circuit

3x AND gates
2x OR gates



AB	00	01	11	10
00	0	0	1	0
01	0	0	1	1
11	0	0	0	1
10	1	1	0	0

K-map

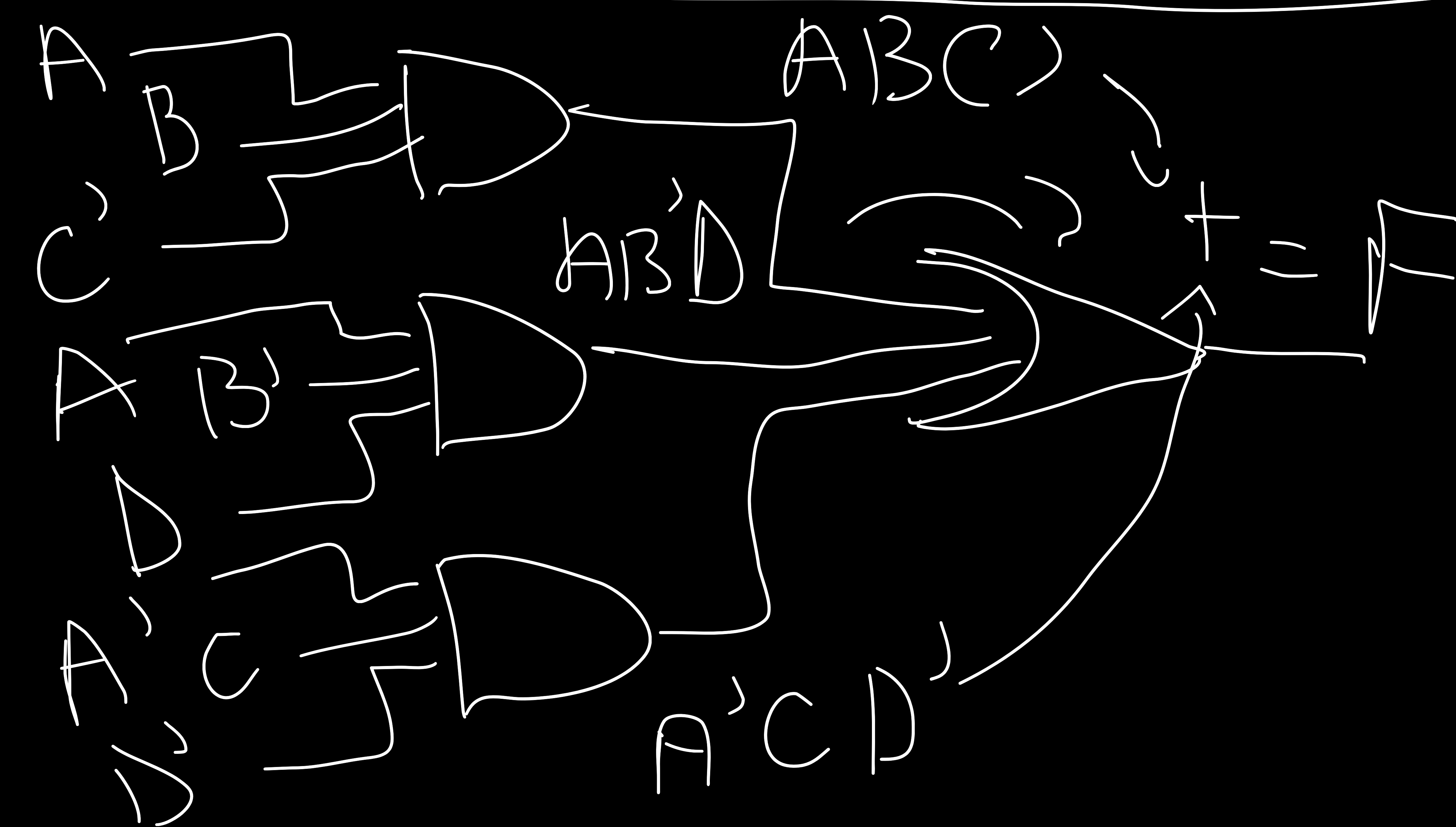
✓

$$F = ABC + AB'D + A'CD$$

↙
Circuit

3 × AND gates

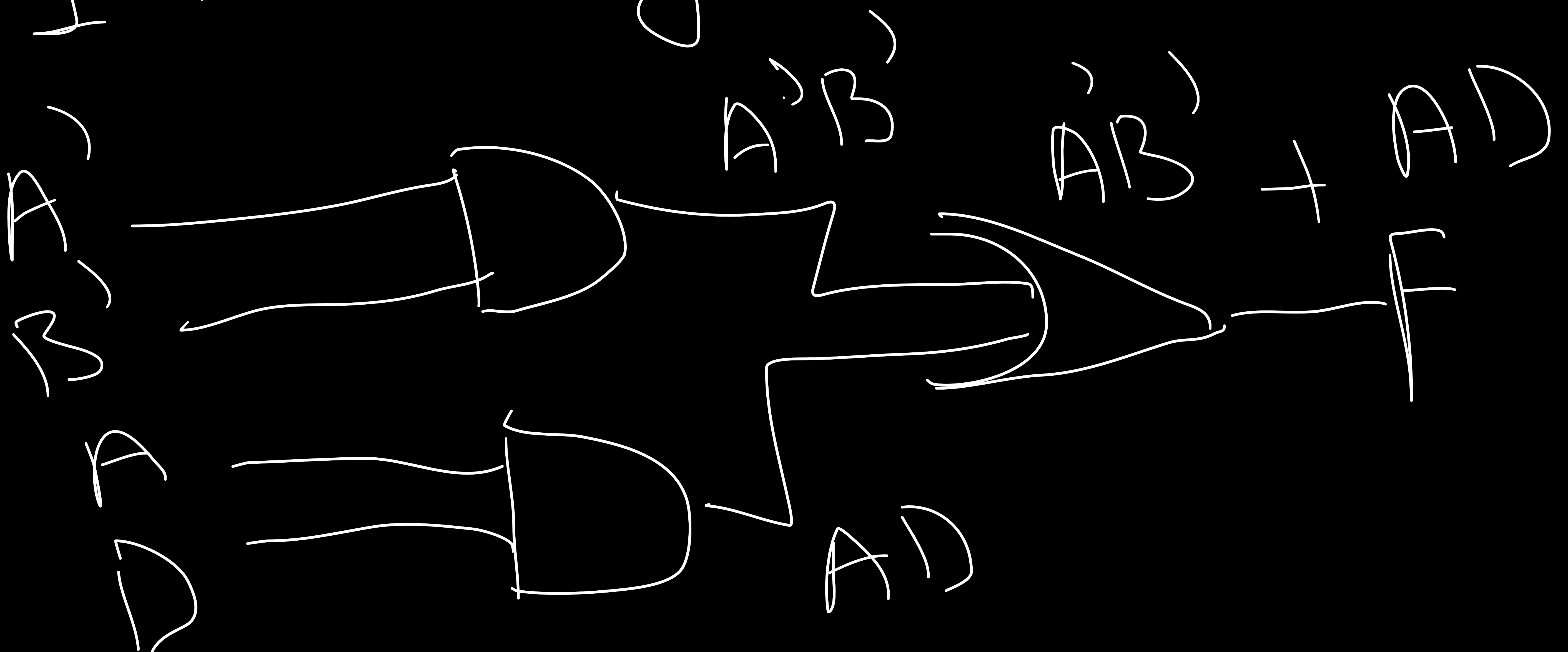
2 × OR gates



AB \ CD		00	01	11	10
CD	00	1	0	0	0
	01	1	0	1	1
	11	1	0	1	1
	10	1	0	0	0

$$F = \overline{A}B + AD$$

2x AND gates
1x OR gates



$$F = AC + AB'$$

$$A = 0 \quad \Rightarrow \quad F = 0$$

$$A = 1$$

AB		00		01		11		10	
C	0	0	0	0	0	0	0	1	1
	1	0	0	1	1	1	1	1	1

$$F = AB' + AC + A'BC$$

AB

C

	00	01	11	10
0	0	0	0	0
1	0	1	1	1

$$F = AB' + A'CD + ABC'$$

AB		00	01	11	10
CD	00	0	0	1	1
	01	0	0	1	1
	11	1	1	0	1
	10	0	0	0	1

$$F = C + AD' + A'BC'D + AB'C$$

AB		00	01	11	10
CD	00	0	0	1	1
	01	0	1	0	0
	11	1	1	1	1
	10	1	1	1	1

